Quadratics: Expanding & Factorising

Practice test 5 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 8x + 6x + 48$ (1) = $\mathbf{x^2 + 14x + 48}$ (1)

A2. $x^2 + 4x - 10x - 40$ (1) = $\mathbf{x^2 - 6x - 40}$ (1)

A3. $(x + 9)(x + 9) = x^2 + 9x + 9x + 81$ (1) = $\mathbf{x^2 + 18x + 81}$ (1)

A4. $10x^2 + 15x - 4x - 6$ (1) = $\mathbf{10x^2 + 11x - 6}$ (1)

Section B — Expanding three brackets

B1. $(x + 4)(x - 3) = x^2 + x - 12$ (1) $(x^2 + x - 12)(x + 1) = x^3 + x^2 + x^2 + x - 12x - 12$ (1) = $\mathbf{x^3 + 2x^2 - 11x - 12}$ (1)

B2. $(2x + 3)(x - 1) = 2x^2 + x - 3$ (1) $(2x^2 + x - 3)(x + 5) = 2x^3 + 10x^2 + x^2 + 5x - 3x - 15$ (1) = $\mathbf{2x^3 + 11x^2 + 2x - 15}$ (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to 24, add to 11: 3 and 8 (1) **(x + 3)(x + 8)** (1)

C2. Multiply to –33, add to –8: –11 and 3 (1) **(x – 11)(x + 3)** (1)

C3. Difference of two squares: **(x + 11)(x – 11)** (1)

C4. $3(x^2 - 9x + 20)$ (1) = **3(x – 4)(x – 5)** (1)

Section D — Factorising $ax^2 + bx + c$

D1. $2 \times 12 = 24$; numbers: 8 and 3, so $2x^2 + 8x + 3x + 12 = 2x(x + 4) + 3(x + 4)$ (1) **(2x + 3)(x + 4)** (1)

D2. $3 \times -6 = -18$; numbers: –9 and 2, so $3x^2 - 9x + 2x - 6 = 3x(x - 3) + 2(x - 3)$ (1) **(3x + 2)(x – 3)** (1)

D3. $3(2x^2 - x - 10)$ (1) = **3(2x – 5)(x + 2)** (1)

Section E — Using factorising

E1. $(x + 8)(x - 5) = 0$ (1) $x + 8 = 0$ or $x - 5 = 0$ (1) **x = –8 or x = 5** (1)

E2. Numerator = $(x + 5)(x - 5)$ (1) denominator = $(x + 5)(x - 3)$ (1) cancel (x + 5): **(x – 5) / (x – 3)** (1)

Section F — Problem solving & proof

F1. (a) $(x - 5)^2 = x^2 - 10x + 25$ (1) so $x^2 - 10x + 18 = \mathbf{(x - 5)^2 - 7}$ (1)
(b) **(5, –7)** (1)

F2. $(n^2 + 6n + 9) - (n^2 + 2n + 1)$ (1) = $4n + 8$ (1) = $4(n + 2)$, which is **4 × an integer, so a multiple of 4** (1)

F3. $(65 + 35)(65 - 35) = 100 \times 30 =$ **3000** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: $(x – 7)(x + 4)$ is the same as $(x + 4)(x – 7)$. “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.